

Module Description Guide

1. Important Points

Name:

"..."

Description:

- What was the structure of the module?
Example: 1.5 h lecture / 1.5 h exercise or additional lab work?
- Which topics are especially important and which are less relevant?
- Why is the module important (or not important) for advanced modules?
- Coursework / exam prerequisites:
What exactly had to be completed?
(Content, workload, time investment, etc.)

Exam:

- What is the format of the assessment?
Example: oral or written exam

If written:

- Calculations / theory – what is the structure?
- How many tasks are there?
- How much time is available?

If oral:

- How is the exam structured?
- What is the goal in the exam?
(e.g. answering as many questions as possible or giving very detailed answers with additional information)

Tips:

- What should students already know beforehand?
- Are there any useful books or resources?
- Which exercises or coursework are especially important?
- Is there anything students should absolutely avoid?
- Are there specific requirements for the module?
(e.g. programming knowledge, software skills, etc.)

2. Example Texts

Technical Drawing

Description

The module consists of 90-minute lectures, exercises, and tutorials where students learn the fundamentals of technical drawing as well as how to draw objects from different perspectives. It ranges from “How to correctly hold a pencil” to “How to represent intersecting bodies.”

As coursework requirements, students must complete hand-drawn assignments as well as computer-based 3D models.

Exam

The exam is written and usually consists of one theory section and two drawing sections.

Tips

- Definitely attend the exercises and tutorials, as they provide the basis for the assignments and are the best place to ask questions.
- Start the assignments early, since they can become very time-consuming, especially if you are perfectionistic.
- The theory and CAD tutorials are important foundations for the module *Design Engineering*.

Design Engineering

Description

This module spans two semesters and consists of 90-minute lectures and exercises, with tutorials only during the first half. Students learn how to identify, draw, model, dimension, and verify the strength of components in technical drawings. The main focus lies on calculations.

Coursework requirements include hand drawings, online quizzes, 3D modeling, and calculations.

Exam

The exam consists of two times three sections:
for each semester there is one theory section, one drawing section, and one calculation section.

Tips

- Definitely attend exercises and tutorials, as they form the basis for the coursework and provide opportunities to ask questions.
 - Start coursework early because some assignments are extensive and very time-consuming, especially the final “complex assignment.”
 - Do not leave the online tests until the last minute, particularly the calculation tests.
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Materials Engineering

Description

The module spans two semesters.

The first semester consists of 90-minute lectures and biweekly exercises, while the second semester focuses on laboratory work.

Students learn about different materials, their chemical structures, and how they can be treated to modify the properties of the final product.

To pass the coursework requirements, students must successfully complete the laboratory sessions, which include both questioning and demonstrations of material testing methods.

Exam

The exam is written.

Practice exams are extremely helpful because some questions are reused in the final exam.

Tips

- Definitely attend the lectures because the lecturer works extensively with board notes that are not included in the slides.
- If lecture recordings from the COVID period are available, use them for studying, as wording and definitions are very important in the exam.
- Prepare thoroughly for the laboratory sessions because students may be questioned in detail there.